Class 12 Mathematics

Chapter 10: Vector Algebra

Practice Set B: step-wise solutions

20 questions **47** marks **3** levels

Each row is one line of working. The red number in the right margin is what an examiner gives for that line, as in the CBSE marking scheme (½ is half a mark). Write every line that carries a mark. ✓ ½

Level 1

8 marks

1 MCQ Board-pattern — 1 mark

The unit vector $\hat{n}$ perpendicular to both $\vec{a} = \hat{i} + \hat{j}$ and $\vec{b} = \hat{j} + \hat{k}$, such that $\vec{a}$, $\vec{b}$, $\hat{n}$ form a right-handed system, is:

(A) $\dfrac{-\hat{i} + \hat{j} - \hat{k}}{\sqrt{3}}$ **(B)** $\dfrac{\hat{i} + \hat{j} + \hat{k}}{\sqrt{3}}$ **(C)** $\hat{i} - \hat{j} + \hat{k}$ **(D)** $\dfrac{\hat{i} - \hat{j} + \hat{k}}{\sqrt{3}}$

Formula / idea: $\vec{a}$, $\vec{b}$, $\vec{a} \times \vec{b}$ form a right-handed system, so $\hat{n} = \dfrac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|}$.

1 For a right-handed system, $\hat{n}$ is along $\vec{a} \times \vec{b}$ (in this order).

$$\hat{n} = \frac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|}$$

2 Find the cross product.

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{vmatrix} = \hat{i} - \hat{j} + \hat{k}$$

3 Divide by its magnitude $\sqrt{1 + 1 + 1} = \sqrt{3}$. ✓ 1

$$\hat{n} = \frac{\hat{i} - \hat{j} + \hat{k}}{\sqrt{3}}$$

4 Hence the answer is option (D).

Answer **(D)** $\dfrac{\hat{i} - \hat{j} + \hat{k}}{\sqrt{3}}$

Watch out: Choosing (A): that is along $\vec{b} \times \vec{a}$, the left-handed order; (C) is not a unit vector.

2 MCQ NCERT Exemplar 1 mark

If $\vec{a}, \vec{b}, \vec{c}$ are three vectors such that $\vec{a}+\vec{b}+\vec{c}=\vec{0}$ and $|\vec{a}|=2$, $|\vec{b}|=3$, $|\vec{c}|=5$, then the value of $\vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{c}+\vec{c}\cdot\vec{a}$ is:

(A) 0 **(B)** 1 **(C)** -19 **(D)** 38

Formula / idea: $|\vec{a}+\vec{b}+\vec{c}|^2=|\vec{a}|^2+|\vec{b}|^2+|\vec{c}|^2+2(\vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{c}+\vec{c}\cdot\vec{a})$.

1 Take the dot product of $\vec{a}+\vec{b}+\vec{c}=\vec{0}$ with itself.
$|\vec{a}+\vec{b}+\vec{c}|^2=0$

2 Expand; let $S=\vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{c}+\vec{c}\cdot\vec{a}$.
$|\vec{a}|^2+|\vec{b}|^2+|\vec{c}|^2+2S=0$

3 Put the magnitudes and solve. ✓ 1
$4+9+25+2S=0 \Rightarrow S=-19$

4 Hence the answer is option (C).

Answer **(C)** -19

Watch out: Choosing (D): dropping the minus sign and the factor 2 in $2S=-38$.

3 MCQ Board-pattern 1 mark

If $\vec{a}=\hat{i}+\lambda\hat{j}+2\hat{k}$ and $\vec{b}=2\hat{i}-\hat{j}+\hat{k}$, then the value(s) of λ for which $\vec{a}+\vec{b}$ is perpendicular to $\vec{a}-\vec{b}$ is/are:

(A) 1 only **(B)** ± 1 **(C)** 4 **(D)** 0

Formula / idea: $(\vec{a}+\vec{b})\cdot(\vec{a}-\vec{b})=|\vec{a}|^2-|\vec{b}|^2$.

1 Perpendicular vectors have dot product zero.
$(\vec{a}+\vec{b})\cdot(\vec{a}-\vec{b})=0$

2 Expand; the terms $\vec{a}\cdot\vec{b}$ and $\vec{b}\cdot\vec{a}$ cancel.
$|\vec{a}|^2-|\vec{b}|^2=0 \Rightarrow |\vec{a}|^2=|\vec{b}|^2$

3 Put the magnitudes. ✓ 1
$1+\lambda^2+4=4+1+1 \Rightarrow \lambda^2=1 \Rightarrow \lambda=\pm 1$

4 Hence the answer is option (B).

Answer **(B)** ± 1

Watch out: Choosing (C): solving $\vec{a}\cdot\vec{b}=0$, which makes $\vec{a}\perp\vec{b}$, not $(\vec{a}+\vec{b})\perp(\vec{a}-\vec{b})$.

4 MCQ Board-pattern 1 mark

If $|\vec{a}| = 2$, $|\vec{b}| = 3$ and $|\vec{a} + \vec{b}| = \sqrt{19}$, then $|\vec{a} \times \vec{b}|$ is equal to:

(A) $3\sqrt{3}$ **(B)** 3 **(C)** 27 **(D)** 6

Formula / idea: $|\vec{a} \times \vec{b}|^2 + (\vec{a} \cdot \vec{b})^2 = |\vec{a}|^2|\vec{b}|^2$.

1 Expand $|\vec{a} + \vec{b}|^2$ to get $\vec{a} \cdot \vec{b}$.
$19 = 4 + 9 + 2\,\vec{a} \cdot \vec{b} \Rightarrow \vec{a} \cdot \vec{b} = 3$

2 Use the identity for the cross product.
$|\vec{a} \times \vec{b}|^2 = |\vec{a}|^2|\vec{b}|^2 - (\vec{a} \cdot \vec{b})^2 = 36 - 9 = 27$

3 Take the square root. ✓ 1
$|\vec{a} \times \vec{b}| = \sqrt{27} = 3\sqrt{3}$

4 Hence the answer is option (A).

Answer **(A)** $3\sqrt{3}$

Watch out: Choosing (C): stopping at $|\vec{a} \times \vec{b}|^2 = 27$; or (B): giving $\vec{a} \cdot \vec{b}$ instead.

5 MCQ NCERT Exemplar 1 mark

If A, B, C, D are the points with position vectors $\hat{i} + \hat{j} - \hat{k}, 2\hat{i} - \hat{j} + 3\hat{k}, 2\hat{i} - 3\hat{k}, 3\hat{i} - 2\hat{j} + \hat{k}$ respectively, then the projection of $\overrightarrow{AB}$ along $\overrightarrow{CD}$ is:

(A) $\frac{1}{\sqrt{21}}$ **(B)** 21 **(C)** $\sqrt{21}$ **(D)** $-\sqrt{21}$

Formula / idea: Projection of $\vec{a}$ along $\vec{b} = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|}$.

1 Find both vectors as head minus tail.
$\overrightarrow{AB} = \hat{i} - 2\hat{j} + 4\hat{k}, \quad \overrightarrow{CD} = \hat{i} - 2\hat{j} + 4\hat{k}$

2 Find the dot product and $|\overrightarrow{CD}|$.
$\overrightarrow{AB} \cdot \overrightarrow{CD} = 1 + 4 + 16 = 21, \quad |\overrightarrow{CD}| = \sqrt{21}$

3 Divide by $|\overrightarrow{CD}|$. ✓ 1
$\text{Projection} = \frac{21}{\sqrt{21}} = \sqrt{21}$

4 Hence the answer is option (C).

Answer **(C)** $\sqrt{21}$

Watch out: Choosing (B): forgetting to divide $\overrightarrow{AB} \cdot \overrightarrow{CD}$ by $|\overrightarrow{CD}|$.

6 **Assertion-Reason** Board-pattern 1 mark

Assertion (A): The vectors $\vec{a} = 2\hat{i} - 3\hat{j} + 4\hat{k}$ and $\vec{b} = -4\hat{i} + 6\hat{j} - 8\hat{k}$ are collinear. **Reason (R):** Two collinear vectors always have the same direction.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A).

(C) **Assertion (A) is true, but Reason (R) is false.**

(D) Assertion (A) is false, but Reason (R) is true.

Formula / idea: $\vec{a}, \vec{b}$ are collinear if $\vec{b} = \lambda\vec{a}$; $\lambda > 0$ gives like vectors, $\lambda < 0$ unlike vectors.

1 Compare $\vec{b}$ with $\vec{a}$.
$\vec{b} = -2(2\hat{i} - 3\hat{j} + 4\hat{k}) = -2\vec{a}$

2 So $\vec{b} = \lambda\vec{a}$ with $\lambda = -2$; the vectors are collinear. **A is true.**

3 Since $\lambda < 0$, $\vec{b}$ points opposite to $\vec{a}$. **R is false.**
Collinear vectors may be like or unlike.

4 Hence the correct option is (C). ✓ 1

Answer **(C) A is true, but R is false**

Watch out: Thinking collinear means same direction; collinear (parallel) vectors can point in opposite directions.

7 **Short answer** NCERT Exemplar 2 marks

Find the area of the parallelogram whose diagonals are $2\hat{i} - \hat{j} + \hat{k}$ and $\hat{i} + 3\hat{j} - \hat{k}$.

Formula / idea: Area of a parallelogram with diagonals $\vec{d}_1, \vec{d}_2 = \frac{1}{2}|\vec{d}_1 \times \vec{d}_2|$.

1 Write the formula for diagonals. ✓ ½
$\text{Area} = \frac{1}{2}|\vec{d}_1 \times \vec{d}_2|$
Adjacent sides are $\frac{1}{2}(\vec{d}_1 + \vec{d}_2)$ and $\frac{1}{2}(\vec{d}_1 - \vec{d}_2)$.

2 Find the cross product. ✓ ½
$\vec{d}_1 \times \vec{d}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -1 & 1 \\ 1 & 3 & -1 \end{vmatrix} = -2\hat{i} + 3\hat{j} + 7\hat{k}$

3 Find its magnitude. ✓ ½
$|\vec{d}_1 \times \vec{d}_2| = \sqrt{4 + 9 + 49} = \sqrt{62}$

4 Halve it. Hence the required area is

$$\text{Area} = \frac{\sqrt{62}}{2} \text{ sq units}$$

✓ ½

Answer **Area** $= \frac{\sqrt{62}}{2}$ **sq units**

Watch out: Not halving: $|\vec{d_1} \times \vec{d_2}|$ is the formula for adjacent sides, not diagonals.

Level 2

16 marks

8 **Short answer** NCERT Exemplar 2 marks

A vector $\vec{r}$ has magnitude 14 and direction ratios $2, 3, -6$. Find the direction cosines and components of $\vec{r}$, given that $\vec{r}$ makes an acute angle with the x-axis.

Formula / idea: $l = \frac{a}{\sqrt{a^2+b^2+c^2}}$, etc.; $\vec{r} = |\vec{r}|(l\hat{i} + m\hat{j} + n\hat{k})$.

1 Find the root of the sum of squares of the direction ratios.

$$\sqrt{2^2 + 3^2 + (-6)^2} = \sqrt{49} = 7$$

✓ ½

2 Take the $+$ sign, since the angle with the x-axis is acute.

$$l = \frac{2}{7}, \quad m = \frac{3}{7}, \quad n = -\frac{6}{7}$$

Acute angle means $l = \cos\alpha > 0$.

✓ ½

3 Multiply the unit vector by the magnitude 14.

$$\vec{r} = 14\left(\frac{2}{7}\hat{i} + \frac{3}{7}\hat{j} - \frac{6}{7}\hat{k}\right) = 4\hat{i} + 6\hat{j} - 12\hat{k}$$

✓ ½

4 State the components.

scalar components: $4,\ 6,\ -12$; vector components: $4\hat{i},\ 6\hat{j},\ -12\hat{k}$

✓ ½

Answer **d.c.** $\frac{2}{7}, \frac{3}{7}, -\frac{6}{7}$**;** $\vec{r} = 4\hat{i} + 6\hat{j} - 12\hat{k}$**, components** $4, 6, -12$

Watch out: Taking $-\frac{2}{7}, -\frac{3}{7}, \frac{6}{7}$: these make an obtuse angle with the x-axis.

9 **Short answer** NCERT Exemplar 2 marks

Find a vector of magnitude 6, which is perpendicular to both the vectors $2\hat{i} - \hat{j} + 2\hat{k}$ and $4\hat{i} - \hat{j} + 3\hat{k}$.

Formula / idea: $\vec{a} \times \vec{b}$ is perpendicular to both $\vec{a}$ and $\vec{b}$.

1 Let $\vec{a} = 2\hat{i} - \hat{j} + 2\hat{k}$, $\vec{b} = 4\hat{i} - \hat{j} + 3\hat{k}$. Find $\vec{a} \times \vec{b}$. ✓ ½

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -1 & 2 \\ 4 & -1 & 3 \end{vmatrix} = -\hat{i} + 2\hat{j} + 2\hat{k}$$

2 Find its magnitude. ✓ ½

$$|\vec{a} \times \vec{b}| = \sqrt{1 + 4 + 4} = 3$$

3 Unit vectors perpendicular to both. ✓ ½

$$\hat{n} = \pm\frac{1}{3}(-\hat{i} + 2\hat{j} + 2\hat{k})$$

4 Multiply by 6. Hence the required vectors are ✓ ½

$$6\hat{n} = \pm 2(-\hat{i} + 2\hat{j} + 2\hat{k}) = \pm(-2\hat{i} + 4\hat{j} + 4\hat{k})$$

Answer $\pm(-2\hat{i} + 4\hat{j} + 4\hat{k})$

Watch out: Giving only one sign; $-(\vec{a} \times \vec{b})$ is also perpendicular to both vectors.

10 **Short answer** NCERT Exemplar 2 marks

The vectors $\vec{a} = 3\hat{i} - 2\hat{j} + 2\hat{k}$ and $\vec{b} = -\hat{i} - 2\hat{k}$ are the adjacent sides of a parallelogram. Find the acute angle between its diagonals.

Formula / idea: Diagonals are $\vec{a} + \vec{b}$ and $\vec{a} - \vec{b}$; $\cos\theta = \dfrac{\vec{d_1} \cdot \vec{d_2}}{|\vec{d_1}||\vec{d_2}|}$.

1 The diagonals are the sum and the difference of the sides. ✓ ½

$$\vec{d_1} = \vec{a} + \vec{b} = 2\hat{i} - 2\hat{j} + 0\hat{k}, \quad \vec{d_2} = \vec{a} - \vec{b} = 4\hat{i} - 2\hat{j} + 4\hat{k}$$

2 Find the dot product and the magnitudes. ✓ ½

$$\vec{d_1} \cdot \vec{d_2} = 8 + 4 + 0 = 12, \quad |\vec{d_1}| = 2\sqrt{2}, \quad |\vec{d_2}| = 6$$

3 Use the angle formula. ✓ ½

$$\cos\theta = \frac{12}{2\sqrt{2} \times 6} = \frac{1}{\sqrt{2}}$$

4 Hence the acute angle between the diagonals is
$\theta = \frac{\pi}{4}$ ✓ ½

Answer $\frac{\pi}{4}$

Watch out: Finding the angle between the sides $\vec{a}$ and $\vec{b}$ instead of between the diagonals.

11 **Short answer** Board-pattern 3 marks

Using vectors, find the area of the triangle with vertices $A(1, 0, 2)$, $B(2, 4, 1)$ and $C(3, 2, 3)$. Hence find the length of the perpendicular from A to BC.

Formula / idea: Area of $\triangle ABC = \frac{1}{2}|\overrightarrow{AB} \times \overrightarrow{AC}| = \frac{1}{2} \times BC \times h$.

1 Find two sides from A.
$\overrightarrow{AB} = \hat{i} + 4\hat{j} - \hat{k}, \quad \overrightarrow{AC} = 2\hat{i} + 2\hat{j} + \hat{k}$ ✓ ½

2 Find the cross product.
$\overrightarrow{AB} \times \overrightarrow{AC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 4 & -1 \\ 2 & 2 & 1 \end{vmatrix} = 6\hat{i} - 3\hat{j} - 6\hat{k}$ ✓ ½

3 Find its magnitude.
$|\overrightarrow{AB} \times \overrightarrow{AC}| = \sqrt{36 + 9 + 36} = 9$ ✓ ½

4 The area is half of this.
$\text{Area} = \frac{9}{2}$ sq units ✓ ½

5 Find the base BC.
$\overrightarrow{BC} = \hat{i} - 2\hat{j} + 2\hat{k}, \quad |\overrightarrow{BC}| = \sqrt{1 + 4 + 4} = 3$ ✓ ½

6 Use area $= \frac{1}{2} \times$ base $\times$ height.
$\frac{9}{2} = \frac{1}{2} \times 3 \times h \Rightarrow h = 3$ units ✓ ½

Answer **Area $= \frac{9}{2}$ sq units; perpendicular from A to $BC = 3$ units**

Watch out: Taking $|\overrightarrow{AB} \times \overrightarrow{AC}| = 9$ as the area, which gives $h = 6$; the triangle is half of it.

12 **Short answer** PYQ-style 3 marks

Find λ if the projection of $\vec{a} = \lambda\hat{i} + \hat{j} + 4\hat{k}$ on $\vec{b} = 2\hat{i} + 6\hat{j} + 3\hat{k}$ is 4 units. Hence find the projection vector of $\vec{a}$ on $\vec{b}$.

Formula / idea: Projection of $\vec{a}$ on $\vec{b} = \dfrac{\vec{a}\cdot\vec{b}}{|\vec{b}|}$; projection vector $= \left(\dfrac{\vec{a}\cdot\vec{b}}{|\vec{b}|}\right)\hat{b}$.

1	Write the formula. $\text{Projection of } \vec{a} \text{ on } \vec{b} = \dfrac{\vec{a}\cdot\vec{b}}{\lvert\vec{b}\rvert}$	✓ ½
2	Find $\vec{a}\cdot\vec{b}$ and $\lvert\vec{b}\rvert$. $\vec{a}\cdot\vec{b} = 2\lambda + 6 + 12 = 2\lambda + 18, \quad \lvert\vec{b}\rvert = \sqrt{4+36+9} = 7$	✓ ½
3	Set the projection equal to 4 and solve. $\dfrac{2\lambda+18}{7} = 4 \Rightarrow 2\lambda = 10 \Rightarrow \lambda = 5$	✓ 1
4	Projection vector = projection $\times$ unit vector along $\vec{b}$. $4\,\hat{b} = 4\cdot\dfrac{2\hat{i}+6\hat{j}+3\hat{k}}{7}$	✓ ½
5	Simplify. $\text{Projection vector} = \dfrac{8}{7}\hat{i} + \dfrac{24}{7}\hat{j} + \dfrac{12}{7}\hat{k}$	✓ ½

Answer $\lambda = 5$; **projection vector** $= \dfrac{8}{7}\hat{i} + \dfrac{24}{7}\hat{j} + \dfrac{12}{7}\hat{k}$

Watch out: Writing the projection vector as $4\vec{b}$; multiply 4 by the unit vector $\hat{b}$, not by $\vec{b}$.

13 **Short answer** Board-pattern 2 marks

If $|\vec{a}| = 2$, $|\vec{b}| = 1$ and the angle between $\vec{a}$ and $\vec{b}$ is $\dfrac{\pi}{3}$, find the value of λ for which $\vec{a} + \lambda\vec{b}$ is perpendicular to $\vec{a} - 3\vec{b}$.

Formula / idea: $\vec{u} \perp \vec{v} \iff \vec{u}\cdot\vec{v} = 0$; $\vec{a}\cdot\vec{b} = |\vec{a}||\vec{b}|\cos\theta$.

1	Find $\vec{a}\cdot\vec{b}$. $\vec{a}\cdot\vec{b} = 2 \times 1 \times \cos\dfrac{\pi}{3} = 1$	✓ ½
2	Perpendicular means the dot product is zero. $(\vec{a} + \lambda\vec{b})\cdot(\vec{a} - 3\vec{b}) = 0$	✓ ½
3	Expand the dot product. $\lvert\vec{a}\rvert^2 - 3\,\vec{a}\cdot\vec{b} + \lambda\,\vec{b}\cdot\vec{a} - 3\lambda\lvert\vec{b}\rvert^2 = 0$	✓ ½

4 Substitute the values and solve. ✓ ½

$$4 - 3 + \lambda - 3\lambda = 0 \Rightarrow 1 - 2\lambda = 0 \Rightarrow \lambda = \frac{1}{2}$$

Answer $\lambda = \frac{1}{2}$

Watch out: Taking $\vec{a} \cdot \vec{b} = 0$; the vectors are at $\frac{\pi}{3}$, so $\vec{a} \cdot \vec{b} = 1$.

14 **Short answer** NCERT Exemplar 2 marks

If $\vec{a}$ and $\vec{b}$ are the position vectors of A and B respectively, find the position vector of a point C in BA produced such that $BC = 1.5\,BA$.

Formula / idea: $\overrightarrow{PQ}$ = (p.v. of Q) − (p.v. of P).

1 C lies on BA produced, so $\overrightarrow{BC}$ has the direction of $\overrightarrow{BA}$. ✓ ½

$$\overrightarrow{BC} = 1.5\,\overrightarrow{BA} = \frac{3}{2}\overrightarrow{BA}$$

2 Write both sides with position vectors; let $\vec{c}$ be the p.v. of C. ✓ ½

$$\vec{c} - \vec{b} = \frac{3}{2}(\vec{a} - \vec{b})$$

3 Solve for $\vec{c}$. ✓ ½

$$\vec{c} = \frac{3}{2}\vec{a} - \frac{3}{2}\vec{b} + \vec{b} = \frac{3}{2}\vec{a} - \frac{1}{2}\vec{b}$$

4 Hence the position vector of C is ✓ ½

$$\overrightarrow{OC} = \frac{3\vec{a} - \vec{b}}{2}$$

Same as dividing BA externally in the ratio $3 : 1$.

Answer $\overrightarrow{OC} = \frac{3\vec{a} - \vec{b}}{2}$

Watch out: Using $\overrightarrow{AB}$ instead of $\overrightarrow{BA}$; that puts C beyond B, on the wrong side.

Level 3

23 marks

15 **Short answer** NCERT Exemplar 3 marks

If $\vec{a} + \vec{b} + \vec{c} = \vec{0}$, show that $\vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a}$. Interpret the result geometrically.

Formula / idea: $\vec{a} \times \vec{a} = \vec{0}$ and $\vec{a} \times \vec{b} = -\vec{b} \times \vec{a}$.

1 Take the cross product of $\vec{a}$ with both sides. ✓ ½

$$\vec{a} \times (\vec{a} + \vec{b} + \vec{c}) = \vec{a} \times \vec{0} = \vec{0}$$

2	Expand and use $\vec{a}\times\vec{a}=\vec{0}$. $\vec{a}\times\vec{b}+\vec{a}\times\vec{c}=\vec{0}\ \Rightarrow\ \vec{a}\times\vec{b}=-\vec{a}\times\vec{c}=\vec{c}\times\vec{a}$	✓ ½
3	Similarly, take the cross product of $\vec{b}$ with both sides. $\vec{b}\times\vec{a}+\vec{b}\times\vec{c}=\vec{0}\ \Rightarrow\ \vec{b}\times\vec{c}=-\vec{b}\times\vec{a}=\vec{a}\times\vec{b}$	✓ ½
4	Combine the two results. $\vec{a}\times\vec{b}=\vec{b}\times\vec{c}=\vec{c}\times\vec{a}$	✓ ½
5	Geometrically, $\vec{a}+\vec{b}+\vec{c}=\vec{0}$ means $\vec{a},\vec{b},\vec{c}$ are the sides of a triangle taken in order.	✓ ½
6	Each cross product has magnitude twice the area of this same triangle. $\lvert\vec{a}\times\vec{b}\rvert=\lvert\vec{b}\times\vec{c}\rvert=\lvert\vec{c}\times\vec{a}\rvert=2\times(\text{area of the triangle})$ All three are also normal to the plane of the triangle.	✓ ½

Answer $\vec{a}\times\vec{b}=\vec{b}\times\vec{c}=\vec{c}\times\vec{a}$**; each has magnitude twice the area of the triangle formed by** $\vec{a},\vec{b},\vec{c}$**.**

Watch out: Writing $\vec{a}\times\vec{c}=\vec{c}\times\vec{a}$; changing the order changes the sign.

16 **Long answer** NCERT Exemplar 5 marks

If $\vec{a},\vec{b},\vec{c}$ determine the vertices of a triangle, show that $\frac{1}{2}\left[\vec{b}\times\vec{c}+\vec{c}\times\vec{a}+\vec{a}\times\vec{b}\right]$ gives the vector area of the triangle. Hence deduce the condition that the three points $\vec{a}$, $\vec{b}$, $\vec{c}$ are collinear. Also find the unit vector normal to the plane of the triangle.

Formula / idea: Vector area of $\triangle ABC=\frac{1}{2}\overrightarrow{AB}\times\overrightarrow{AC}$: its magnitude is the area, its direction is normal to the plane.

1	Let A, B, C have position vectors $\vec{a}$, $\vec{b}$, $\vec{c}$. Write two sides. $\overrightarrow{AB}=\vec{b}-\vec{a},\quad\overrightarrow{AC}=\vec{c}-\vec{a}$	✓ ½
2	Write the vector area of $\triangle ABC$. $\vec{\Delta}=\frac{1}{2}\overrightarrow{AB}\times\overrightarrow{AC}=\frac{1}{2}(\vec{b}-\vec{a})\times(\vec{c}-\vec{a})$	✓ ½
3	Expand the cross product term by term. $(\vec{b}-\vec{a})\times(\vec{c}-\vec{a})=\vec{b}\times\vec{c}-\vec{b}\times\vec{a}-\vec{a}\times\vec{c}+\vec{a}\times\vec{a}$	✓ 1
4	Use $\vec{a}\times\vec{a}=\vec{0}$, $-\vec{b}\times\vec{a}=\vec{a}\times\vec{b}$ and $-\vec{a}\times\vec{c}=\vec{c}\times\vec{a}$. $(\vec{b}-\vec{a})\times(\vec{c}-\vec{a})=\vec{b}\times\vec{c}+\vec{c}\times\vec{a}+\vec{a}\times\vec{b}$	✓ ½
5	Hence the vector area of the triangle is $\vec{\Delta}=\frac{1}{2}\left[\vec{b}\times\vec{c}+\vec{c}\times\vec{a}+\vec{a}\times\vec{b}\right]$	✓ ½
6	Three points are collinear exactly when the triangle has zero area, i.e. $\vec{\Delta}=\vec{0}$. $\vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a}=\vec{0}$	✓ 1

7 $\vec{\Delta}$ is perpendicular to the plane of the triangle; divide it by its magnitude. ✓ 1

$$\hat{n} = \frac{\vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a}}{|\vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a}|}$$

Answer **Vector area** $= \frac{1}{2}[\vec{b}\times\vec{c}+\vec{c}\times\vec{a}+\vec{a}\times\vec{b}]$ **(shown); collinear** $\iff \vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a}=\vec{0}$**;** $\hat{n} = \dfrac{\vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a}}{|\vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a}|}$

Watch out: Losing a sign when swapping the order: $\vec{b}\times\vec{a} = -\vec{a}\times\vec{b}$, not $+\vec{a}\times\vec{b}$.

17 **Short answer** Board-pattern 3 marks

Find a vector of magnitude 9 units which is perpendicular to both $\vec{a} = 2\hat{i}+\hat{j}+2\hat{k}$ and $\vec{b} = 3\hat{i}+2\hat{j}+2\hat{k}$, and which makes an obtuse angle with the y-axis.

Formula / idea: Vectors of magnitude m perpendicular to $\vec{a}$ and $\vec{b}$ are $\pm m\dfrac{\vec{a}\times\vec{b}}{|\vec{a}\times\vec{b}|}$.

1 $\vec{a}\times\vec{b}$ is perpendicular to both; find it. ✓ 1

$$\vec{a}\times\vec{b} = \begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 2 & 1 & 2\\ 3 & 2 & 2\end{vmatrix} = -2\hat{i}+2\hat{j}+\hat{k}$$

2 Find its magnitude. ✓ ½

$$|\vec{a}\times\vec{b}| = \sqrt{4+4+1} = 3$$

3 Vectors of magnitude 9 perpendicular to both. ✓ ½

$$\pm\frac{9}{3}(-2\hat{i}+2\hat{j}+\hat{k}) = \pm(-6\hat{i}+6\hat{j}+3\hat{k})$$

4 An obtuse angle with the y-axis needs a negative $\hat{j}$-component. ✓ ½

$\cos\beta = \dfrac{\hat{j}\text{-component}}{|\vec{r}|} < 0$ for an obtuse β.

5 So take the minus sign. ✓ ½

$$\vec{r} = 6\hat{i}-6\hat{j}-3\hat{k}$$

Answer $6\hat{i}-6\hat{j}-3\hat{k}$

Watch out: Giving both signs; the obtuse-angle condition keeps only one of them.

18 **Short answer** Board-pattern 3 marks

$ABCD$ is a plane quadrilateral with vertices $A(1,1,1)$, $B(3,1,3)$, $C(3,4,6)$ and $D(1,2,2)$. Using vectors, find its area by dividing it along the diagonal AC.

Formula / idea: $\text{ar}(ABCD) = \text{ar}(\triangle ABC) + \text{ar}(\triangle ACD)$, each $= \frac{1}{2}|\text{cross product of two sides}|$.

1 Find the vectors from A. ✓ ½

$\overrightarrow{AB} = 2\hat{i} + 0\hat{j} + 2\hat{k}, \quad \overrightarrow{AC} = 2\hat{i} + 3\hat{j} + 5\hat{k}, \quad \overrightarrow{AD} = 0\hat{i} + \hat{j} + \hat{k}$

2 For $\triangle ABC$, find $\overrightarrow{AB} \times \overrightarrow{AC}$. ✓ ½

$$\overrightarrow{AB} \times \overrightarrow{AC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 0 & 2 \\ 2 & 3 & 5 \end{vmatrix} = -6\hat{i} - 6\hat{j} + 6\hat{k}$$

3 Halve its magnitude. ✓ ½

$$\text{ar}(\triangle ABC) = \frac{1}{2}\sqrt{36 + 36 + 36} = \frac{6\sqrt{3}}{2} = 3\sqrt{3}$$

4 For $\triangle ACD$, find $\overrightarrow{AC} \times \overrightarrow{AD}$. ✓ ½

$$\overrightarrow{AC} \times \overrightarrow{AD} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 5 \\ 0 & 1 & 1 \end{vmatrix} = -2\hat{i} - 2\hat{j} + 2\hat{k}$$

5 Halve its magnitude. ✓ ½

$$\text{ar}(\triangle ACD) = \frac{1}{2}\sqrt{4 + 4 + 4} = \frac{2\sqrt{3}}{2} = \sqrt{3}$$

6 Add the two areas. ✓ ½

$\text{ar}(ABCD) = 3\sqrt{3} + \sqrt{3} = 4\sqrt{3}$ sq units

Check: $\frac{1}{2}|\overrightarrow{AC} \times \overrightarrow{BD}| = 4\sqrt{3}$ as well.

Answer **Area $= 4\sqrt{3}$ sq units**

Watch out: Using $|\overrightarrow{AB} \times \overrightarrow{AD}|$ as for a parallelogram; here $\overrightarrow{BC} = 3\overrightarrow{AD}$, so $ABCD$ is a trapezium.

19 **Long answer** Board-pattern 5 marks

In a triangle ABC, let a, b, c be the lengths of the sides opposite to the vertices A, B, C respectively.

(i) Using vectors, prove that $\cos A = \dfrac{b^2 + c^2 - a^2}{2bc}$. 3 marks

(ii) Hence find $\cos A$ for the triangle with vertices $A(1,2,1)$, $B(3,4,2)$ and $C(5,2,-2)$, and check your answer using $\overrightarrow{AB} \cdot \overrightarrow{AC}$. 2 marks

Formula / idea: $|\vec{u}-\vec{v}|^2 = |\vec{u}|^2 + |\vec{v}|^2 - 2\,\vec{u}\cdot\vec{v}$ and $\vec{u}\cdot\vec{v} = |\vec{u}||\vec{v}|\cos\theta$.

(i) 3 marks

1	Let $\vec{b} = \overrightarrow{AC}$ and $\vec{c} = \overrightarrow{AB}$; the angle between them is A. $\lvert\vec{b}\rvert = b, \quad \lvert\vec{c}\rvert = c, \quad \vec{b}\cdot\vec{c} = bc\cos A$	✓ ½
2	By the triangle law, $\overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AC}$. $\overrightarrow{BC} = \vec{b} - \vec{c}$	✓ ½
3	The length of BC is a; square it. $a^2 = \lvert\vec{b}-\vec{c}\rvert^2 = (\vec{b}-\vec{c})\cdot(\vec{b}-\vec{c})$	✓ ½
4	Expand the dot product. $a^2 = \lvert\vec{b}\rvert^2 + \lvert\vec{c}\rvert^2 - 2\,\vec{b}\cdot\vec{c} = b^2 + c^2 - 2bc\cos A$	✓ 1
5	Rearrange for $\cos A$. $\cos A = \dfrac{b^2 + c^2 - a^2}{2bc}$	✓ ½

Answer (i) $\cos A = \dfrac{b^2 + c^2 - a^2}{2bc}$ **(proved)**

(ii) 2 marks

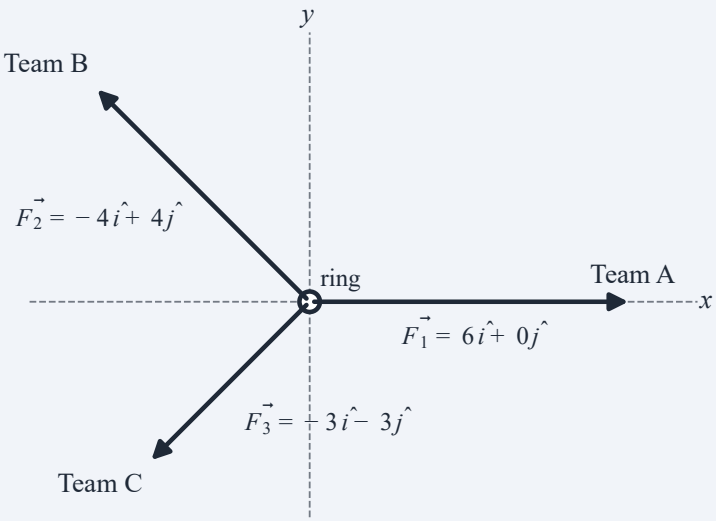
1	Find the side vectors. $\overrightarrow{AB} = 2\hat{i} + 2\hat{j} + \hat{k}, \quad \overrightarrow{AC} = 4\hat{i} + 0\hat{j} - 3\hat{k}, \quad \overrightarrow{BC} = 2\hat{i} - 2\hat{j} - 4\hat{k}$	✓ ½
2	Find the side lengths. $c = AB = 3, \quad b = AC = 5, \quad a = BC = \sqrt{24} = 2\sqrt{6}$	✓ ½
3	Use the result of part (i). $\cos A = \dfrac{25 + 9 - 24}{2 \times 5 \times 3} = \dfrac{10}{30} = \dfrac{1}{3}$	✓ ½
4	Check with the dot product of the two sides from A. $\cos A = \dfrac{\overrightarrow{AB}\cdot\overrightarrow{AC}}{\lvert\overrightarrow{AB}\rvert\lvert\overrightarrow{AC}\rvert} = \dfrac{8 + 0 - 3}{3 \times 5} = \dfrac{1}{3}$ ✓	✓ ½

Answer (ii) $\cos A = \dfrac{1}{3}$

Watch out: For angle A both vectors must start at A; using $\overrightarrow{BA}$ with $\overrightarrow{AC}$ flips the sign of $\cos A$.

20 **Case study** CBSE SQP 2023-24 4 marks

Case Study: Teams A, B, C went for playing a tug of war game. Teams A, B, C have attached a rope to a metal ring and each is trying to pull the ring into its own area. Team A pulls with force $\vec{F}_1 = 6\hat{i} + 0\hat{j}$ kN, Team B pulls with force $\vec{F}_2 = -4\hat{i} + 4\hat{j}$ kN and Team C pulls with force $\vec{F}_3 = -3\hat{i} - 3\hat{j}$ kN.



(i) What is the magnitude of the force of Team A? 1 mark

(ii) Which team will win the game? 1 mark

(iii) In what direction is the ring getting pulled? 2 marks

Formula / idea: The ring moves along the resultant $\vec{F}_1 + \vec{F}_2 + \vec{F}_3$; $|x\hat{i} + y\hat{j}| = \sqrt{x^2 + y^2}$.

(i) 1 mark

1 Find the magnitude of $\vec{F}_1$.
$|\vec{F}_1| = \sqrt{6^2 + 0^2} = 6$ kN ✓ 1

Answer (i) 6 **kN**

(ii) 1 mark

1 Add the pulls of teams A and C.
$\vec{F}_1 + \vec{F}_3 = 3\hat{i} - 3\hat{j} = 3(\hat{i} - \hat{j})$

2 Write $\vec{F}_2$ in the same form; it is opposite to $\vec{F}_1 + \vec{F}_3$.
$\vec{F}_2 = -4(\hat{i} - \hat{j})$ ✓ ½

3 Compare the magnitudes.
$|\vec{F}_2| = 4\sqrt{2} > |\vec{F}_1 + \vec{F}_3| = 3\sqrt{2}$

4 So the ring moves towards B. Hence Team B wins. ✓ ½

Answer (ii) **Team** B

(iii) 2 marks

1	Find the resultant force. $\vec{F} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 = (6 - 4 - 3)\hat{i} + (0 + 4 - 3)\hat{j} = -\hat{i} + \hat{j}$	✓ 1
2	Let θ be its angle with the positive x-axis. $\cos\theta = \frac{-1}{\sqrt{2}}, \quad \sin\theta = \frac{1}{\sqrt{2}}$	✓ ½
3	So θ is in the second quadrant; the ring is pulled along $-\hat{i} + \hat{j}$. $\theta = \pi - \frac{\pi}{4} = \frac{3\pi}{4}$	✓ ½

Answer (iii) **Along $\frac{-\hat{i} + \hat{j}}{\sqrt{2}}$, i.e. at $\frac{3\pi}{4}$ with the positive x-axis**

Watch out: Picking Team A for its largest single pull; A and C partly cancel, so compare $\vec{F}_2$ with $\vec{F}_1 + \vec{F}_3$.